A

GLA University, Mathura

End-Term Examination, 2012-13

Course: - B.Tech. **II-Year, III-Sem.**

Subject: - Mathematics-III (AHM-201) **Uni. Roll No:-**

Time:-2Hour 30 Minutes **Total Marks:-40**

Notes:-

1.Attempt **ALL** questions of Section- A, **ANY FOUR** from Section- B, and **ANY FOUR** from Section- C.
2.All parts of a question (a, b, etc.) should be answered at one place.
3.Answer should be brief and to-the-point and be supplemented with neat sketches.
4.Any missing or wrong data may be assumed suitably giving proper justification.

Section - A (1 × 10 = 10)

Q .1 The nature of singularity of $f(z)=\sin\frac{1}{z-1}$ at $z=1$ is

* Removable * Isolated essential * Non Isolated essential * Non of these

Q.2 The value of $\int \frac{dz}{z^2+9}$, where C is $|z-3i|=4$ is

* $\frac{\pi}{3}$ * $-\frac{\pi}{3}$ * 0 * $-\frac{\pi}{4}$

Q.3 If $f(x)=x\sin x$ in $(0,2\pi)$, then value of a_0 is

* 0 $f(x)=x\sin x$ * -1 * 2 * -2

Q.4 The equation whose roots are decrease by 2 to the roots of the eqn. $x^3-6x^2+12x-8=0$ is

* $x^3-x^2-8=0$ * $x^3-8=0$ * $x^3-3x^2+3x-1=0$ * $x^3=0$

Q.5 The function $f(z)=e^z$ has an isolated essential singularity at

* 0 * -1 * $\frac{\pi}{3}$ * None of these

Q.6 the Cauchy Reimann equations in polar form are

* $ru_r=v_\theta, rv_r=-u_\theta$ * $u_r=rv_\theta, rv_r=-u_\theta$ * $v_\theta=\frac{1}{r}u_r, u_\theta=-\frac{1}{r}v_r$ * $ru_r=v_\theta, rv_r=u_\theta$

Q.8 The Z- transforms of $\delta(k-n)$ is

* z^n * $z^{1/n}$ * $1/z^n$ * z

Q.8 The Fourier sine transforms of e^{-x} is

* $\frac{p}{p^2-1}$ * $\frac{p}{1-p^2}$ * $\frac{p}{p^2+1}$ * $\frac{1}{p}$

Q.9 Using the fact that $\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$, $\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2}$ is equal to

* $\frac{\pi^2}{12}$ * $\frac{\pi^2}{12}-1$ * $\frac{\pi^2}{8}$ * $\frac{\pi^2}{8}-1$

Q.10 Let $f: C\to C$ be a complex function of the form $f=u(x,y)+iv(x,y)$ and suppose that $u(x,y)=3x^2y$, then

* f cannot be analytic for any v * f is analytic for suitable v * f is analytic for all v * none of these

Section - B (2.5 × 4 = 10)

Q1. Solve the biquadratic equation $x^4 - 3x^2 - 42x - 40 = 0$ by using Ferrari's method.

Q.2 Express $f(x) = \pi x - x^2$ as a half – range sine series in the interval $(0, \pi)$ up to first three terms.

Q.3 Find the inverse Fourier transform of $f(p) = e^{-|p|y}$.

Q.4 Solve the following equation by Z- transform:

$y_{x+2} - 2\cos\alpha\ y_{x+1} + y_x = 0,\ y_0 = 0,\ y_1 = 1.$

Q.5 Find p such that the function f(z) expressed in the polar coordinates as

$f(z) = r^2\cos 2\theta + i\, r^2 \sin p\theta$ is analytic.

Section - C (5 × 4 = 20)

Q.1 Find the values of c_1 and c_2 such that the function

$f(z) = x^2 + c_1 y^2 - 2xy + i(c_2 x^2 - y^2 + 2xy)$ is analytic.

Q.2 Show that the function $u = e^{-2xy}\sin(x^2 - y^2)$ is harmonic. Find the conjugate function v and express $u + iv$ as an analytic function of z.

Q.3 Determine poles and residue at each pole and hence evaluate the integral $\int_c \frac{3z^2 + z}{z^2 - 1} dz$, where c is the circle $|z| = 2$.

Q.4 Find the Laurent's series which represent the function $f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$ when

(a) $2 < |z| < 3$, and (b) $|z| > 3$

Q.5 By residue theorem, evaluate:

$$\int_0^{2\pi} \frac{\cos 2\theta}{5 + 4\cos\theta} d\theta$$

GLA University C

End Term Exam [ODD] 2012-13

B. Tech: II Year, 3th semester [C.S, E.C.]

Ethics & Values (AHE: 201)

Marks: 40 Time: 2.30 Hours

This paper consists of four passages. Carefully read the passages and answer the multiple choices questions. Mark-distribution is given in front of the passages.

Passage: 1 (1X6=6 Marks)

Utilitarianism suits to human nature as a moral theory to decide a right choice out of wrongs. In this moral theory, we make a choice on the basis of maximum utility, and a Utilitarian defines maximum utility in terms of the results or the consequences an action produces. A Utilitarian argues that a right choice/action/decision is something that produces maximum good for maximum number of people. But, they do not care about preferences and choices of an individual. However, a Utilitarian takes care of a society as a whole. Jeremy Bentham, in his work on Utilitarianism, defines that by the very human nature we desire pleasure and avoid pain. And, whatever provides pleasure over pain is preferable as a right choice for Bentham to act on. This is a naturalistic way of defining the rightness of a choice or an action. Although John Stuart Mill follows Bentham and is agreed with him on the most of the points of Utilitarianism, Mill makes a distinction between lower order and higher order pleasure. He says a higher order pleasure is something that involves human capacities of thinking, differentiating, and judgment and lower order pleasure is simply we enjoy pleasure without testing the nature of pleasure on the grounds of human rational capacities. Here, we can understand that though Mill gives preference to pleasure and happiness but he gives more importance to rational features of human beings in making qualitative choices and differences amongst the pleasures of different kinds. For an example, a drug addict who derives pleasure in the addiction of drugs only minds the quantity of pleasure s/he gets through it. S/he does not rationalize the consequences or the result of drug addiction. However, the reformative and rehabilitative programs running across the whole world against the drug addiction are working on the suggestions of Mill that we should rationalize the consequences or results in terms of quality of pleasures. Once this qualitative difference making capacity is being realized, a drug addict can get rid of addiction and can live qualitatively a better life.

Further, Mill develops an interesting point on the idea of freedom that freedom is something that cannot be assessed on the grounds of utility. In other words, he argues that the criteria of defining the right things to do on the grounds of maximum good of maximum number is fine to the extent it does not affect ones fundamental rights say freedom. But, if a maximum good of maximum number is overriding individual rights then Mill argues he would not go with Utilitarianism. He says; if it comes to chose between the Utilitarianism and Individual Freedom, I would choose individual freedom over the utilitarianism. Let's understand this argument by a prevailing problem of Indian culture namely, Honor killing. The issue of Honor Killing involves

a conflict between individual choice and social sanctioning (Social legitimacy). If we analyze this problem on the views of Jeremy Bantam then we would say Honor Killing is justified since it takes care of the preferences of maximum number (Particular society) in order to define the right thing to do. However, Mill would object Honor Killing because here maximum good is overriding individual rights and freedom. In this way, we find Mill more reasonable in selecting the quality of pleasure and happiness and defining utility of that accordingly with the help of reason. This is what we can say rise of reason for the moral judgment to decide what is right and what is wrong.

Questions:

1) Mill would object the Honor Killing because:
   - Mill is more reasonable in selecting the quality of pleasure.
   - Mill defines utility of pleasure or happiness by reason.
   - In honor killing maximum good overrides individual freedom.
   - None of them.
2) A utilitarian defines the right thing:
   - On the basis of individual choice.
   - On the basis of results or consequences.
   - On the basis of maximum utility.
   - On the basis of maximum good for maximum number.
3) Jeremy Bentham and Mill both are agree on these following points:
   - A qualitatively better pleasure produces more utility than a qualitatively inferior pleasure.
   - Right thing to do is something that produces maximum utility.
   - Humans desire happiness by nature.
   - Utilitarianism suits to human nature.
4) Mill makes a difference between higher order pleasure and lower order pleasure. He prefers to have higher order pleasure because;
   - It involves intentions and intuitions.
   - It involves rational capacities of a human being.
   - It involves thinking and judgment.
   - It involves differentiation.
5) Mill's utilitarianism slightly differs from that of Bentham's utilitarianism. Which of the statement is/are not correct about Mill's utilitarianism:
   - Mill is more reasonable in choosing the pleasures.
   - Mill makes no distinction amongst the different kinds of pleasures.
   - Mill would chose maximum good over individual rights.
   - Mill gives preference to individual freedom over maximum utility.
6) Anti Drug addiction rehabilitation programs running across the whole world would agree with John Stuart Mill one the following points:
   - We should go for pleasures only.
   - We should rationalize the results or consequences of our pleasures that we get through drug-addiction.
   - We should make a qualitative difference in choosing our pleasures.
   - We should choose those pleasures which make our life qualitatively better.

Passage: 2 (1X6=6 Marks)

Euthanasia (assisted suicide) has been a long debated issue in Ethics. Euthanasia, a morally baffled issue, needs a clarification, that morality of euthanasia is being tested on the following points: a) a person has been suffering from a terminal (incurable) disease; b) a person has been suffering from physical and psychological pains, c) The death of a person is quite predictable, and d) the person, undergoing through such pains, wants to die with the assistance of other.

However, we have to understand an important point that euthanasia differs from that of suicide in a way that in the case of a suicide a person do not want to live, while in euthanasia a person wants to die. In other words, when a person commits suicide s/he does not want to live because of personal life issues and socio-psychological problems. But, in euthanasia somebody does not find herself/ himself to be capable of to live the life. For instance somebody is suffering from a terminal physical disease and is on the life support system. However, there are three fundamental questions underlying the debate. First, do we have right to take assisted suicide? Second, is mercy killing justified? Third, is moral sanctioning of assisted suicide consistent with humanitarian spirit? Such questions are really tough to answer. However, we have got two theoretical points of views to discuss this issue: 1) Libertarianism, and 2) Kantianism. Libertarians argue that we own ourselves in the way that an individual has got complete freedom to oneself to decide in what manner one wants to live one's life. Since we possess ourselves we have freedom to decide even on the question of deciding for assisted suicide. Libertarians would sanction euthanasia on the grounds that an individual has got complete rights on his/her own life. And, if one's decision does not harm the other then one is free to take any decision. However, Kantians unlike Libertarians would consider the euthanasia on the ground of rationality. Although like Libertarian Kantians advocate the rights of individuals and admit that we own ourselves and one must be given complete rights of decision making to oneself, Kantians maintain that one's decision making to oneself must be governed by reasons. Kantian morality is deontological in its nature. Deontological system of morality is a system in which the moral worth of rules, duties and responsibilities is being examined on the basis of the very nature of those rules, duties and morals. In this system, morality is viewed as end in itself. Hence, Kantians would argue that the complete freedom of oneself to decide for oneself is fine but in case of euthanasia if we demand for assisted suicide or mercy killing then we actually treat life as a means to achieve some other ends like good status, happiness, and success etc... Therefore, Kantians would not sanction euthanasia because for them life is valuable in itself; an end in itself that cannot be treated as a means to fulfill some other ends. Kantians are more rational on this issue because they apply wisdom that one must view life as an end in itself in order to save the spirit of humanity.

Questions:

1) Euthanasia differs from suicide because:
   - In Euthanasia a person is free to decide on the issue of taking one's life and in suicide a person is not free to do so.
   - In case of Euthanasia a person wants to die, while in suicide a person does not want to live.
   - In Euthanasia a person is not capable to live.
   - Somebody is suffering from a terminal disease and is on life support system.
2) Euthanasia is morally confused and debated issue. It involves the following points:
   - A person has been feeling physical and psychological pains.
   - A person is suffering from an incurable disease.
   - Death of a person is quite predictable.
   - A person wants to die with his or her own assistance.

do not lose temper and establish harmony. Finally, we move to understand the idea of Communicative Stance. It refers to the argumentative position of a communicator. In order to develop good communication ability one has to develop a position on the bases of reasons and arguments. The position we take up in a communicative situation is called our communicative stance. It is the way one presents or offers a certain reasons, explanations or clarifications to advocate, defend or deny certain things. Hence, having a sound communications skills does not simply mean that one can speak in English rather it means that one can take a communicative stance when one speaks. Moreover, we need to be consistent to our position; it means what we develop and present through reasons and arguments during communication at one point we must not be inconsistent or contradict those reasons and arguments at other, otherwise we will be in self-contradicting position in a communicative situation. And, if it happens then it is a sign of poor communication. Therefore, the mantra of Communicative Stance is: Go with reasons and be consistent.

Questions:

1) Standard communication involves three basic ideas:
   - Structure of ideas
   - Structure of life
   - Forms of life
   - Forms of Language
2) Having a sound Communicative Stance means:
   - One accepts the other's positions.
   - One is communicating on the grounds of reasons.
   - One develops a position by offering reasons and arguments.
   - One is reasonable and consistent.
3) What are signs of having poor Linguistic Psyche?
   - One speaks loudly.
   - One is nervous and unconfident in a communicative situation.
   - One's psychological fear is affecting one's communication.
   - One's heartbeat goes faster and lips get dry.
4) Why does a University need to have a good culture?
   - In order to restrict the freedom of students within the frontiers of rules and regulations.
   - In order to have a good academic environment.
   - In order to improve the personalities of students.
   - In order to provide students the right ways to develop their academic and personal lives.
5) What are the grounds to make a judgment whether a student is cultured or not?
   - The way a student fulfills his/her self-interests.
   - The way a student behaves in his/her academic life.
   - The way a student participates in various activities and communicates with others.
   - The way a student approaches to his/her professional and person life.
6) What do we mean by soft-skills in communication?
   - Communicating without losing the temperament and maintaining harmony in communicative situations.
   - Communicating in a polite and decent manner.
   - Communicating without getting into fighting mode or in a clashing situation.
7) Having a sound Linguistic Psyche means:
   - Communicating with others in an authoritative manner.

- Communicating in a confident and comfortable manner.
- Internalizing and realizing the communication and be relaxed and free.
- Communicating with prepared mind and having clear ideas and intentions.

8) A second order , comprehensive, analysis of communication involves the following points:

- One's impression on others.
- One's gestures & postures.
- One's ideas & intentions.
- One's psychological involvement and whole body language in a communicative situation.

9) The problems of communication are not only centered to the uses of language and grammar but also are closely connected with some more basic characteristics namely;

- Communicative Interests
- Linguistic Psyche
- Soft-Skills
- Communicative Stance

Passage: 4 (2X5=10 Marks)

These days we witness a drastic change in the approaches and attitude of students. The changes are very much perceptible in their forms and ways of life. And, rationally we need to accept it because it's not a new phenomenon; it always occurs in a span of time. However, we need to understand one thing that there are certain skills which have been crucial for a student life regardless the fact that changes occur as the time changes. Although the skills we are going to discuss and analyze are not specific to student life, they are developed and solidified in student life. As student life provides a platform to nurture and cement those necessary skills. A good student life blessed with the necessary skills pushes a student further to accomplish certain developments in his/her professional and personal life. A student becomes rationally sound and confident in order to handle the difficult and problematic situations of life. And, once this problem solving personality coupled with rationally critical attitude is being lived in oneself one can count himself/herself as a developed and advanced student. However, what are those necessary skills which make this realization possible in a student? The skills are as follow: 1) Rationality, 2) Critical Thinking, 3) Decision Making, and 4) Problem Solving. Rationality is a faculty of our mind by which we understand and explain the events and situations of life on the grounds of reasons. And, when we do not do that or choose irrational grounds we cannot improve and develop. For example, suppose there are two students A and B. Both of the students have got poor grades in their exams. When they reflect why it happened with them. A, by choosing irrational ground, understands that it happened because s/he has been suffering through a bad luck or God is not happy with her/him. On the other hand, B, by choosing rational ground, understands that it happened because s/he was not serious in her/his studies, and s/he did not make adequate efforts to get good grades. Hence, first of all, we must try to find out the reasons in the very event and not the outside of the event. In this way, the door of improvements and developments opens with rational attitude. However, Critical Thinking is a step ahead to rationality because in thinking critical we make a painstaking effort in order to examine our rationality and remove all that is false and irrational by crosschecking and counter questioning. Having done critical thinking what remain (the true facts and reasons) becomes practically rational to pursue and practice. For instance, we should not react in life on the basis of others comment and ideas as we do in case of rumors and become a victim of someone's

dirty trick or political mind. Hence, lack of critical thinking can bring setbacks and distract us from the development. Once you have sound and solid base, after critical thinking, to act upon certain issues of life then comes the third skill in the picture, i.e. the capacity of Decision Making. Decision making involves your value judgment on given choices at a time. Although most of times we have to make a decision in which we have to figure out the right thing out of wrong and complex choices, sometimes even seemingly right choices carry a thin line difference there we have to pick up a real right choice out of the choices which appeared to be right only. Finally, the Problem-Solving is not an independent skill but it naturally comes when a person acquires the first three skills (Rationality, Critical Thinking, and Decision Making). However, we need to adopt Problem-solving attitude and abstain ourselves from being Problem-creating attitude. For instance, some people when they get into certain problems they rationalize and think critically and thereafter try to find out the resolutions following a good decision making, while some others who are having problem creating attitude they simple try to fulfill their ego and anger. They don't rationalize and think critically on certain issues and create problems for others. That's why I always say to my students to make a qualitative difference between the two statements: 1) I am in mud, and 2) I want to be in the mud.

Questions:

1) Why does a student require some basic skill?
   - ➢ In order to become powerful and wealthy.
   - ➢ In order to critically rationalize his/her life issues.
   - ➢ In order to handle difficult and problematic situations of life.
   - ➢ In order to accomplish certain developments for both his/her professional and personal life.
2) Critical-Thinking though based on rationality but qualitatively differs from it because:
   - ➢ Rationality comes after critical thinking.
   - ➢ It examines rationality.
   - ➢ In this process we remove all that is false and irrational.
   - ➢ In this process we crosscheck and counter the rationality.
3) Which of the following statement is not true about Rationality?
   - ➢ Being rational we don't do what is irrational.
   - ➢ Rationality is a faculty of our mind.
   - ➢ Being rational we explain and understand the events of life on the bases of reasons.
   - ➢ Rationality follows one's ego and emotions.
4) "One is in mud and one wants to be in the mud". The statement refers to the following points:
   - ➢ Someone is driven by irrationality fulfills his/her ego and anger.
   - ➢ Someone does not have problem solving attitude.
   - ➢ Someone does not rationalize and think critically.
   - ➢ Someone only makes his/her life easy by creating problems.
5) A sound Decision-Making involves:
   - ➢ A decision which serve one's interests.
   - ➢ Rationality and Critical-Thinking
   - ➢ Picking up a right choice out of the choices which appeared to be right only.
   - ➢ A decision in which we have to figure out the right thing out of wrong and complex choices.

SET B

GLA University, Mathura

End Term Examination, 2012 – 2013

Course: - B. Tech. (CS) II Year, III Semester

Subject: Data Structures and Algorithms (CSE211) **University Roll No.**

Time: 2 Hour 30 Minutes **Total Marks: 40**

Notes:

1. **Part A** is **compulsory.** Answer **Any four** from **Part B** and **Any four** from **part C.**
2. All parts of a question should be answered at one place.
3. Answer should be brief-to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably with justification.
5. Figures on the right-hand side margin indicate full marks.

Part A

Attempt all questions. **[0.5 x 20 = 10]**

1. Which of the following data structure can't store the non-homogeneous data elements?
   - Pointers
   - Records
   - Arrays
   - None of these
2. Which of the following statement is false?
   - Arrays are dense lists and static data structure
   - Pointers store the next data element of a list
   - Data elements in linked list need not be stored in adjacent space in memory
   - Linked lists are collection of the nodes that contain information part and next pointer
3. A variable P is called pointer if
   - P contains the address of an element in DATA
   - P can store only memory addresses
   - P contain the DATA and the address of DATA
   - P points to the address of first element in DATA
4. Which of the following is not the required condition for binary search algorithm?
   - There should be the direct access to the middle element in any sub list
   - The list must be sorted
   - There must be mechanism to delete and/or insert elements in list
   - None of these
5. What is the time complexity of Heap sort?
   - O(n)
   - $O(n\log_2 n)$
   - $O(\log_2 n)$
   - $O(n^2)$
6. Binary search algorithm cannot be applied to
   - Sorted linked list
   - Sorted binary trees
   - Sorted linear array
   - Pointer array

7. In static hashing
   - The hash table size does not increase in any condition
   - The hash table size increases on collision
   - Both can be possible
   - Cannot predict
8. The memory address of the first element of an array is called
   - Floor address
   - Foundation address
   - First address
   - Base address
9. Which of the following algorithm is used for determining single-source shortest path in a weighted digraph (where some of the edge weights may be negative)
   - Dijkstra algorithm
   - Floyd-Warshall algorithm
   - Bellman-Ford algorithm
   - None of these
10. Identify the data structure which allows deletions at both ends of the list but insertion at only one end
    - Output-restricted deque
    - Priority queues
    - Input-restricted deque
    - None of these
11. A binary tree whose every node has either zero or two children is called
    - Complete binary tree
    - Binary search tree
    - Extended binary tree
    - None of these
12. The depth of a complete binary tree is given by
    - $D_n = n\log_2 n+1$
    - $D_n = \log_2 n$
    - $D_n = n\log_2 n$
    - $D_n = \log_2 n+1$
13. When converting binary tree into extended binary tree, all the original nodes in binary tree are
    - External nodes on extended tree
    - Vanished on extended tree
    - Internal nodes on extended tree
    - None of these
14. The post order traversal of a binary tree is DEBFCA. Find out the pre order traversal
    - ADBFEC
    - ABDECF
    - ABFCDE
    - ABDCEF
15. The in order traversal of tree will yield a sorted listing of elements of tree in
    - Binary search trees
    - Heaps
    - Binary trees
    - None of these
16. Reorder the sequence 4 15 8 18 9 11 10 so that when elements are inserted in this new order into an empty binary search tree, the resulting tree has height 3. Recall that by "height" I mean the number of nodes in the longest path from root to leaf.
    - 4 9 8 10 15 11 18
    - 10 8 15 9 4 11 18
    - 10 8 15 4 9 11 18
    - 10 8 4 15 9 11 18
17. The worst case time complexity of Quick sort is:
    - $O(n\log_2 n)$
    - $O(\log_2 n)$
    - $O(n)$
    - $O(n^2)$
18. Assume that a data fie has an index consisting of N items, where N is large. If a binary search of the index is used to find an item, then, of the following, which best approximates the mean number of comparisons required to locate a specified index entry?
    - $N(N+1)/2$
    - $\log_2 N - 1$
    - $(N+1)/2$
    - $N \log_2 N$

19. A queue data structure observes first in first out flow
    - The statement is true if queue is implemented with array but not when queue is implemented as linked list
    - The statement is true for non circular queues only
    - The statement is false as queue observes last in first out flow
    - The statement is true in all conditions
20. A tree is an acyclic graph
    - The statement is incorrect
    - The statement is correct
    - One cannot define one data structure in terms of other data structure so both the options are meaningless
    - None of these

Part B

Attempt any four questions. **[2.5 x 4 = 10]**

1. Given pointers to two nodes in a doubly linked list, write pseudo code for swapping the order of the elements.
2. Show with examples the insertion and deletion of elements in an array based and linked list based stack. What are key differences in both of these implementations?
3. Show the AVL tree inserting keys 3, 9, 5, 6, 1, 2, 11 and 10 in the given order.
4. Show the selection sort steps for given elements 13, 21, 53, 34, 93, 12, 121 and 100 to be sorted in ascending order.
5. What is the minimum height and maximum height of a binary tree with 32 elements? Why?

Part C

Attempt any four questions. **[5 x 4 = 20]**

1. Write algorithm for Merge sort or Quick sort. Do both of these have similar time complexity in Big O notation? Explain your answer.
2. Show with examples the insertion of 10 elements of your choice and then deletion of 4 elements in a B+ tree. Explain the terms overflow and underflow in the context of B+ tree.
3. Write a details note on storage media file organization.
4. Show the Min heap with keys 3, 9, 5, 6, 1, 2, 11 and 10 being inserted in the given order.
5. Show the bubble sort steps for given elements 33, 29, 55, 46, 11, 72, 141 and 10 to be sorted in ascending order.

Univ. Roll No-.....

SET-C

GLA UNIVERSITY,MATHURA

End- Term Exam 2012-13

COURSE: B.TECH. YEAR: II SEM: III Branch: CS

Sub: Object Oriented Programming (ITE 211)

Time: - 2 hour 30 Minutes **Max. Marks:-40**

1.Answer all questions from Group A (which is) **compulsory**, **Any four** from Group B and **Any four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group- A

Write most appropriate answers to the following questions. (20x0.5=10 Marks)

I. A java program used to create a dynamic web page on the browser with help of HTML
- Servelet
- Bean
- Applet
- Swing

II.A package is a collection of
- Classes
- Interfaces
- Classes and Interfaces
- Objects

III. **Which of the following is not a keyword**
- extends
- protected
- implements
- NULL

IV. **In class diagram attributes are shown in**
- Fourth compartment
- Third compartment
- Second compartment
- First compartment

V. **Multiplicity ``1..*'' represents**
- Exactly one
- One or more
- Many
- Zero or more

VI. Which of the following inheritance is not supported by java
- Hierarchical
- Multiple
- Multilevel
- Single Level

VII. UML is a
- Natural language
- Scripting Language
- Modeling Language
- Programming Language

VIII. Which features of the OOP showing the essentials details and hiding the non essential details.
- Inheritance
- Abstraction
- Polymorphism
- Encapsulation

IX. Which of the following is not a keyword in java
- final
- static
- abstract
- override

X. Which of the following is not a feature of java
- distributed
- dynamic
- multithreaded
- platform dependent

XI. The process of building new classes from existing one is called _____.
- Polymorphism
- Structure
- Inheritance
- Cascading

XII. A property which is not true for classes is that they
- are removed from memory when not in use.
- permit data to be hidden from other classes.
- bring together all aspects of an entity in one place.
- Can closely model objects in the real world.

XIII. Additional information sent when an exception is thrown may be placed in
- an object of the exception class.
- the function that caused the error.
- The catch block.
- The throw keyword.

XIV. How many constructors can a class have?

- 0
- 1
- 2
- any number

XV.The new operator

- returns a pointer to the variable
- creates a variable called new
- obtains memory for a new variable
- tells how much memory is available

XVI. Consider the following statements:

int x = 22,y=15;
x = (x>y) ? (x+y) : (x-y);
What will be the value of x after executing these statements?

- 22
- 7
- Error. Cannot be executed
- 37

XVII. A class defined within another class is:

- Nested class
- Inheritance
- Containership
- Encapsulation

XVIII. A copy constructor takes

- no argument
- one argument
- two arguments
- arbitrary no. of arguments

XIX. Which can be passed as an argument to a function?

- constant
- expression
- another function
- all of the above.

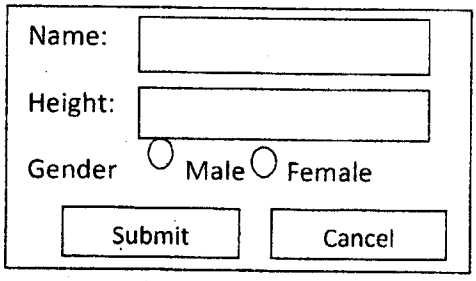
XX. Which of the following may be part of a class definition?

- instance variables
- instance methods
- constructors
- all of the above

Group B

Short Answer Type Questions(Attempt any four) **(2.5x4=10 Marks)**

Q1. What do you mean by Use case? Explain the different elements of Use Case with suitable example.
Q2. Bring out the features that constitute the JAVA language.
Q3.Discuss in detail about creating and accessing a package in JAVA with suitable example.
Q4.How do applets differ from application programs? Under what circumstances it would be required to use applet.
Q5.WAP in java to design the following window through AWT package.



Group C

Long Answer Type Questions (Attempt any Four) **(5x4=20) Marks**

Q1. What is JDBC? Briefly discusses its architecture and all type of drivers.

Or

Discuss basic steps in JDBC and Write a program in JAVA to display the result of table student (StudentID,Name,Address).

Q2.How multithreading is handled in JAVA and what are the different ways to achieve that? Explain with example.
Q3.What is java Bean. Discuss the various steps involved in creating java Beans?
Q4.What do you mean by Servlet and explain the difference between Servlet and CGI.
Q5. Define modeling in UML and it advantages. What are the different views in UML?

GLA University, Mathura
END Term Examination, 2012-13

SET-A

Course: - B.Tech (CS) **II-Year, III-Sem.**
Subject: - Computer Organization (CSE 212) **Uni. Roll No:-**
Time:-2 hours 30 Minutes **Total Marks:-40**

10 Copy
R-1028
R-1156

Notes:-

1) *Answer all questions; Section A is **compulsory**, **Any four** from Section B and **Any four** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer): **(20*0.5 = 10)**

I. $(A0C)_{16}$ is equivalent to
- $(3572)_{10}$
- $(1010\ 0000\ 1100)_2$
- $(1023)_4$
- $(2345)_8$

II. The average time required to reach a storage location in memory and obtain its contents is called the
- seek time
- access time
- turnaround time
- transfer time

III. Which register keeps track of the next instructions stored in memory.
- AR (Address Register)
- PC (Program Counter)
- XR (Index Register)
- AC (Accumulator)

IV. A group of bits that tell the computer to perform a specific operation is known as
- Instruction code
- Micro-operation
- Accumulator
- Register

V. A Stack-organized Computer uses _____ instruction format.
- Indirect addressing
- Zero addressing
- Two-addressing
- Index addressing

VI. How many address lines are needed to address each memory locations in a memory chip of size 2048 x 4?
- 11
- 12
- 10
- 22

VII. If RAM memory chip size is 256 x 1 bits then how many chips required to made 1K bytes of memory ?
- 24
- 12
- 32
- 8

VIII. Which memory is fastest memory in memory organization?
- Hard disk
- Cache memory
- Main memory
- Register

IX. Which is not the part of micro programmed control unit?
- Memory data register
- Control memory
- Control address register
- Micro-program sequencer

X. In immediate addressing the operand is placed
- after OP code in the instruction
- in memory
- in the CPU register
- in stack

XI. The communication bus which is used to fetch the address of an instruction from memory is called the
- Address Bus
- Data Bus
- Fetch Bus
- Control Bus

XII. Content of the program counter is added to the address part of the instruction in order to obtain the effective address is called.
- Relative address mode.
- Register mode.
- Index addressing mode.
- Implied mode.

XIII. SIMD represents an organization that _____.
- Refers to a computer system capable of processing several programs at the same time.
- Represents organization of single computer containing a control unit, processor unit and a memory unit.
- Includes many processing units under the supervision of a common control unit
- None of the above.

XIV. In computers, subtraction is generally carried out by
- 9's complement
- 1's complement
- 10's complement
- 2's complement

XV. Which is represent the condition control statement by two register transfer statements with control functions.
If (P=1) then R1←R2 else if (Q=1) then R1←R3

- P': R1 ←R2 , P'Q: R1 ←R3
- P': R1 ←R2 , P'Q': R1 ← R3
- P: R1 ←R2 , P'Q': R1 ← R3
- P: R1 ←R2 , P'Q: R1 ← R3

XVI. The value of bias in IEEE754 single precision is

- 1023
- 1
- 127
- 0

XVII. Average access time of two level memory hierarchy is 12ns. The Main memory has a hit rate of 90% and access time of 5ns. What is the access time of cache memory?

- 7.5ns
- 750ns
- 75ns
- 9

XVIII. Let SP=00000 in the stack. How many items are there in the stack if Full=0 and Empty=1?

- 64
- 128
- 0
- 32

XIX. Assume a computer with 3-address instruction format, the number of bytes required to encode an instruction is (Assume each address require 24 bits and 1 byte is required for operation code)

- 73
- 10
- 72
- 11

XX. Consider a case where 6 –segment pipeline with clock cycle time 20 ns in each sub operation to execute 25 tasks. Pipeline system will take how much time to complete the tasks?

- 6000ns
- 620ns
- 600ns
- 720ns

SECTION-B (Attempt any four) (4*2.5=10)

I. What do you mean by Flynn's Classification? Discuss Flynn's Classification in brief.

II. Write the difference between Hardwired Control unit and Micro-programmed Control unit. Discuss Micro-programmed control unit in brief.

III. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.
(i) How many selection inputs are there in each multiplexer?
(ii) How many multiplexers are there in the bus? (iii) Specify size of multiplexer.

IV. Write RPN of given expression and write zero-address instruction format for given expression using postfix notation.

$$X = (A - B + C * (D * E / F)) / (G - H + K)$$

V. Write short notes on any two.
(a) Instruction cycle
(b) IEEE 754 floating point representation
(c) Shift operation of data in a register.

SECTION-C (Attempt any four) (4*5=20)

I. A Computer uses RAM chips of 512 x 8 and ROM chips of 2048 x 8. How many RAM chips and ROM chips are needed to provide a RAM memory capacity of 2048 bytes and ROM memory of 2048 bytes? Draw the memory address map for the system and show CPU connection with memory.

II. A virtual memory has an address space 16K words, a memory space has 8K words and block sizes of 2K words. The following reference generated by CPU

4 2 0 1 2 6 1 4 0 1 0 2 3 5 7

Show each page reference change if the replacement algorithm used is (a) FIFO (b) LRU

III. What do you mean by locality of reference? Discuss the different mapping techniques used for cache memory.

IV. What are the different modes of data transfer? Explain the DMA Controller with a block diagram.

V. Why I/O interface is required in I/O organization? Write the difference between Memory- Mapped I/O and Isolated I/O.

GLA University, Mathura **SET-B**

End Term Examination, 2012-13

Course: - B.Tech. (CS) **II-Year, III-Sem.**

Subject: - Discrete Mathematics (CSE-213) **Uni. Roll No:-**

Time:- 150 Minutes **Total Marks:-40**

1) *Answer all questions from Section A, Any four from Section B and Any four from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer) : **(20*0.5 = 10)**

I. Which of the following sets are finite :
A = {seasons in the year}, B = {positive integers less than 1},
C = {odd integers}, D = {positive integral divisors of 12}
• A & D • B & D • A, B & D • C

II. Which of the following sets are equal :
$A = \{x : x^2-4x+3 = 0\}$, $B = \{x : x \in N, x < 3\}$, $C = \{x : x \in N, x \text{ is odd}, x < 5\}$, $D = \{1,3,1\}$
• A & B • A & C • A, C & D • B, C & D

III. Consider the following relations on the set A = {1, 2, 3, 4} and determine which of the relations are reflexive.
$R_1 = \{(1,1), (1,2), (2,3), (1,3), (4,4)\}$ $R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$ $R_3 = A \times A$
• R_2 • R_3 • R_2 & R_3 • R_1

IV. Find the number of relations from A = {a, b, c} to B = {1, 2}
• 12 • 32 • 64 • 6

V. Given that the value of $p \rightarrow q$ is false then determine the value of $(\bar{p} \vee \bar{q}) \rightarrow q$
• F • p • q • T

VI. The generating function for the sequence 1, -2, 3, -4, 5, -6, ... is
• $1/(1-x)$ • $1/(1+x)^2$ • $1/(1-x)^2$ • $1/(1+x)$

VII. Find the number of 3-letter words formed by using A, B, C, D, E, F without repetitions
• 20 • 120 • 216 • 18

VIII. How many 7-letter words can be formed using the letters of the word 'BENZENE'?
• 420 • 840 • 5040 • 210

IX. Find the number of ways that 12 students can be divided into 3 teams so that each team contains 4 students.
• 5775 • 11880 • 34650 • 1320

X. How many committees of five with a given chairperson can be selected from 12 persons?
• 792 • 3960 • 95040 • 330

XI. In how many ways can three boys and two girls sit in a row if the boys and girls are each to sit together?
• 24 • 48 • 120 • 10

XII. A lot contains 12 items of which four are defective. Three items are drawn at random one after the other. Find the probability that all three are nondefective.
• 41/55 • 8/27 • 2/3 • 14/55

XIII. A box contains two white socks and two blue socks. Two socks are drawn at random. Find the probability that they are of the same color?
• 1/4 • 2/3 • 1/3 • 1/2

XIV. The probability that A hits a target is 1/3 and the probability that B hits a target is 1/5. They both fire at a target. Find the probability that neither hits the target?
• 7/15 • 8/15 • 2/5 • 1/15

XV. Among 100 people, at least how many people were born in the same month?
• 8 • 9 • 10 • 7

XVI. What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five grades A, B, C, D, E?
• 24 • 25 • 26 • 21

XVII. If there are 6 vertices in simple graph what is the maximum number of edges in the graph.
• 15 • 30 • 36 • 12

XVIII. How many edges are there in a graph with 20 vertices each of degree 3?
• 15 • 120 • 60 • 30

XIX. If a 6-regular graph has 12 vertices, then determine the total number of edges in the graph?
• 18 • 36 • 72 • 12

XX. Determine the number of edges in a graph with 10 vertices, 6 with degree 4 and 4 with degree 6?
• 24 • 48 • 60 • 12

SECTION – B (Attempt any four) (4*2.5=10)

I. Show that for any two sets A and B, $A - (A \cap B) = A - B$

II. The contrapositive of a statement S is given as "if $x < 2$ then $x+4 < 6$". Write the statement S and its converse.

III. Determine PDNF for following: $(p \wedge \neg q \wedge \neg r) \vee (q \wedge r)$

IV. Determine the validity of the following statement using rules of inference:
"If philosophers are not money-minded and some money-minded persons are not clever then there are some persons who are neither philosophers nor clever."

V. Solve the following recurrence relation :
$a_r - 6\,a_{r-1} + 8a_{r-2} = 4^r$ given $a_0=8$, $a_1=22$

SECTION – C (Attempt any four) (4*5=20)

I. Show that $\langle Z_5, +_5, X_5 \rangle$ is a commutative ring.

II. Consider the operator * defined on Z, the set of integers as: $a*b = a+b+1 \quad \forall a, b \in Z$
Show that (Z, *) is an abelian group.

III. Answer following:
   a. State and prove Lagrange's theorem.
   b. Prove that inverse of each element of a group is unique.

IV. Prove that the sum of degrees of all vertices of a graph is equal to twice the number of edges in graph.
Draw the following graphs:
a. K_4 **b.** $K_{2,3}$ **c.** C_6

V. Let R^+ be the set of all positive real numbers. Show that the function $f : R^+ \rightarrow R$ defined by $f(x) = \log_2(x)$ is a homomorphism of group $(R^+, *)$ to the group $(R, +)$ where * & + are ordinary multiplication & addition respectively and also Prove that (R, +, .) is a commutative ring, iff $\forall x, y \in R$: $(x + y)^2 = x^2 + y^2 + 2xy$.